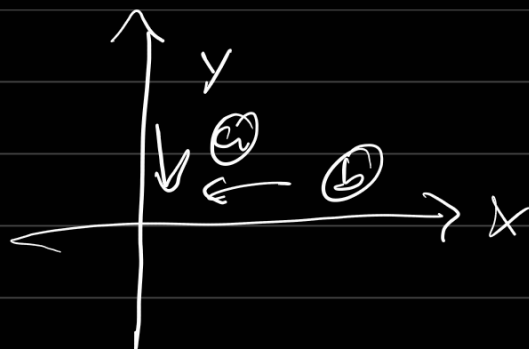


Cauchy Riemann : z

$$\frac{\partial f}{\partial z} = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$



$$z = x + iy, \quad f = u + iv$$

$$u(x, y), v(x, y)$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial z} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial z}$$

$$\boxed{u_x = v_y \quad | \quad u_y = -v_x}$$

$$\frac{\partial u}{\partial x} = (-z) \frac{\partial}{\partial x} (e^{-2zw} \cos(z^2 - w^2))$$

$$+ e^{-2zw} (-1) \sin(z^2 - w^2) \frac{\partial}{\partial x} (z^2 - w^2)$$

$$= -2e^{-2zw} \cos(z^2 - w^2) [u_x w + u w_x]$$

$$- e^{-2zw} \sin(z^2 - w^2) [2u u_x - 2w v_x]$$

$$= -2e^{-2zw} [u_x (\cos(1)w + \sin(1)u) + u_x (\cos(1)u - \sin(1)w)]$$

$$\frac{\partial v}{\partial y} = -2e^{-2zw} \sin(1) [u_y w + u w_y]$$

$$+ e^{-2zw} \cos(1) [2u u_y - 2w v_y]$$

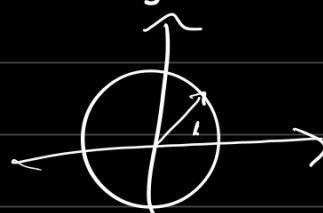
for ∇_y : $u_x = v_y$ and $u_y = -v_x$

$$= -2e^{-2zw} [(\sin(1) [(1-u_x)w + u u_x])$$

$$- \{ \cos(1) (2u(-v_x) - 2w u_x) \}]$$

$$= -2e^{-2zw} \{ \cos(1) [2u v_x + 2u_x w] + \sin(1) (u u_x - v v_x) \}$$

$$I_1 = \oint_{C_1} \frac{\sin(z + \pi/2)}{z} dz$$



$$|z| = 1$$

$$2\pi i \operatorname{Res}(\sin(z + \pi/2))|_{z=z_0} = 2\pi i$$

Find

$$z = re^{i\theta} \Rightarrow dz = dr e^{i\theta} + \dots$$

$$|r| = 1$$

$$z = e^{i\theta} \Rightarrow dz = i e^{i\theta} d\theta = \underline{\underline{i z d\theta}}$$

$$\oint_{C_1} \rightarrow \int_0^{2\pi}$$

$$S_n(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$S_n(z + \pi/2) = \sum_{n=0}^{\infty} \frac{(-1)^n (z + \pi/2)^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n (e^{i\theta} + \pi/2)^{2n+1}}{(2n+1)!}$$

$$S_n(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

$$S_n(z + \pi/2) = \frac{e^{i(z + \pi/2)} - e^{-i(z + \pi/2)}}{2i}$$

$$= \frac{(e^{iz} + e^{-iz})i}{2i} = \cos(z)$$

$$= \int_0^{2\pi} \frac{\cos(z)}{z} i z d\theta$$

$$= \int_0^{2\pi} i \cos(z) d\theta = \int_0^{2\pi} i \frac{e^{iz} + e^{-iz}}{2} d\theta$$

$$= \int_0^{2\pi} i \sum_{n=0}^{\infty} \frac{1}{n!} z^n \left[\frac{i^n + (-i)^n}{2} \right] d\theta$$

$$= \int_0^{2\pi} i \sum_{n=0}^{\infty} \frac{1}{n!} e^{in\theta} i^n \left[\frac{1^n + (-1)^n}{2} \right] d\theta$$

$$= i \int_0^{2\pi} d\theta + i \sum_{n=1}^{\infty} \int_0^{2\pi} \frac{1}{n!} e^{in\theta} i^n [1^n + (-1)^n] d\theta$$

$$= \underline{\underline{2\pi i}} + i \sum \int (1) \left[\frac{1}{n!} e^{in\theta} \right]_0^{2\pi}$$

$$\downarrow e^{i2\pi n} - 1$$

$$\cos(2\pi n) - 1$$

$$(1) - 1 = 0$$

$$= 2\pi i \quad \checkmark$$

$$z=0 \quad \text{and} \quad z=\pi/2$$

$$2\pi i \left[0 + \frac{z}{\pi} \right] = 4i$$



$$|z| = \pi$$

$$\oint \frac{\sin(z)}{z(z-\pi/2)} i z d\theta$$

$$= \oint \frac{i \sin(z)}{z - \pi/2} d\theta$$

$$w \equiv z - \pi/2, \quad dw = dz$$

$$= \oint \frac{i \sin(w + \pi/2)}{w} d\theta$$

$$= \oint \frac{i \cos(w)}{w} d\theta$$

$$= i \int_0^{2\pi} \frac{e^{iw} + e^{-iw}}{2w} d\theta$$

$$=$$

$$z = \pi e^{i\theta} = |w| e^{i\theta} + \pi/2$$

$$\frac{i}{2} \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{1}{n!} w^n i^n [1 + (-1)^n]$$

$$= \frac{i}{2} \int_0^{2\pi} \frac{1}{w} d\theta = \frac{i}{2} \int_0^{2\pi} \frac{1}{|w| e^{i\theta}} d\theta$$

$$= \frac{i}{2|w|} \int_0^{2\pi} e^{-i\theta} d\theta = \frac{i}{2|w|(-i)} (e^{-i\theta} \Big|_0^{2\pi})$$

$$= -\frac{1}{2|w|} \rightarrow 0$$

$$\star \int \frac{\sin(z)}{z(z-\pi/2)} dz$$

$$= \int \frac{\sin(z)}{z-\pi/2} i\pi d\theta$$

$$i\pi e^{i\theta} d\theta$$

$$i\pi z d\theta$$

$$w \equiv z - \pi/2$$

$$|w|e^{i\phi} = \pi e^{i\theta} \pm \pi/2$$

$$i|w|e^{i\phi} d\phi = i\pi e^{i\theta} d\theta$$

$$w d\phi = z d\theta$$

$$d\phi = \frac{z}{z - \pi/2} d\theta$$

$$= \int \frac{\sin(z)}{z} d\phi$$

$$\int \frac{\sin(w + \pi/2)}{w + \pi/2} i\pi d\phi$$

$$w = |w|e^{i\phi}$$

$$\frac{1}{z(z - \pi/2)} = \frac{A}{z} + \frac{B}{z - \pi/2}$$

$$1 = A(z - \pi/2) + Bz$$

$$= z(A + B) - A\frac{\pi}{2} = 1$$

$$A = -B$$

$$-A\frac{\pi}{2} = 1 \Rightarrow A = -\frac{2}{\pi}, B = \frac{2}{\pi}$$

$$\int \sin(z) \left(\frac{2}{\pi} \right) \left[-\frac{1}{z} + \frac{1}{z - \pi/2} \right]$$

$$= \frac{2}{\pi} \int \sin(z) \left(-\frac{1}{z} \right) dz + \frac{2}{\pi} \int \sin(z) \frac{1}{z - \pi/2} dz$$

(a)

(b)

$$(b) \frac{2}{\pi} \int \frac{\sin(z)}{z - \pi/2} dz, w = z - \pi/2, dw = dz$$

$$= \frac{2}{\pi} \int \frac{\sin(w + \pi/2)}{w} dw$$

$$2(2\pi i) = 4\pi i$$

✓

$$= \frac{1}{\pi} (\dots) = 0$$

(i) $z=0$: show

$$\frac{2}{\pi} \int \frac{\sinh(z)}{z} dz$$

$$\sinh(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\frac{2}{\pi} \int \frac{\sinh(z)}{z} i z d\theta = \frac{2}{\pi} \int_0^{2\pi} \sinh(\pi e^{i\theta}) i d\theta$$

$$= \frac{2}{\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{(\pi e^{i\theta})^{2n+1}}{(2n+1)!} (-1)^n$$

$$= \frac{2}{\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} \frac{\pi^{2n+1}}{(2n+1)!} (-1)^n e^{i(2n+1)\theta}$$

$$= \frac{2}{\pi} \sum_{n=0}^{\infty} \frac{\pi^{2n+1}}{(2n+1)!} (-1)^n \left. \frac{1}{i(2n+1)} e^{i(2n+1)\theta} \right|_0^{2\pi}$$

$$= \frac{2}{\pi} \sum_{n=0}^{\infty} \left[e^{i(2n+1)(2\pi)} - 1 \right] (-1)^n$$

$$= 0 \quad \checkmark$$

$$\int \frac{z^{1/2}}{1 + \sqrt{2}z + z^2}$$

$$A = \sqrt{z} - B$$

$$1 + \sqrt{2}z + z^2$$

$$AB = 1$$

$$A + B = \sqrt{2}$$

$$\sqrt{2}B - B^2 = 1$$

$$B = \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

$$* A = \sqrt{2} - \frac{1}{\sqrt{2}} (-1 \pm i)$$

$$= \sqrt{2} \left(1 - \frac{1}{2} (-1 \pm i) \right)$$

$$A = \frac{\sqrt{2}}{2} (1-i)$$

$$+ A = \sqrt{2} \left(\frac{3}{2} - \frac{i}{2} \right)$$

$$B = \frac{\sqrt{2}}{2} (-1+i)$$

$$\beta = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

$$\left[z + \frac{\sqrt{2}}{2}(1-i) \right] \left[z + \frac{\sqrt{2}}{2}(1+i) \right] \quad \checkmark$$

$$\star \frac{1}{2}((1-i)(1+i)) = \frac{1}{2}(1+i-i-i^2) = 1 \quad \checkmark$$

$$\frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}i} - \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}}i + \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} + \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}i}i = \sqrt{2} \quad \checkmark$$

$$\star (z+\alpha)(z+\beta)$$

$$z = -\alpha$$

$$\frac{(-\alpha)^{1/2}}{-\alpha+\beta} + \frac{(-\beta)^{1/2}}{-\beta+\alpha} = i \left(\frac{\alpha^{1/2}}{-\alpha+\beta} - \frac{\beta^{1/2}}{-\alpha+\beta} \right)$$

$$= \frac{i}{-\alpha+\beta} (\alpha^{1/2} - \beta^{1/2})$$

$$\alpha = \frac{\sqrt{2}}{2}(1-i)$$

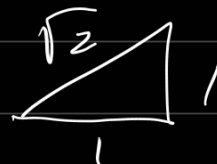
$$-\alpha+\beta = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

$$= \sqrt{2}i$$

$$= \frac{1}{\sqrt{2}} (\alpha^{1/2} - \beta^{1/2})$$

$$\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$



$$\alpha = \frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}}$$

$$= \cos\left(\frac{\pi}{4}\right) - i \sin\left(\frac{\pi}{4}\right)$$

$$= \cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right)$$

$$= e^{-i\pi/4}$$

$$\beta = e^{i\pi/4}$$

$$e^{-i\pi/8}$$

$$- e^{i\pi/8}$$

$$\frac{2\pi i}{\sqrt{2}} (-1) \left(e^{\pi/8} - e^{-\pi/8} \right) \frac{2i}{2i}$$

$$= \frac{4\pi}{\sqrt{2}} \left(\sin\left(\frac{\pi}{8}\right) \right) = 2^{3/2} \pi \sin\left(\frac{\pi}{8}\right)$$

$$z^2 + \sqrt{2} z + 1$$

$$\sqrt{2} \pm \sqrt{2 - 4} / 2 = \frac{\sqrt{2} \pm i\sqrt{2}}{2} = \frac{1}{\sqrt{2}} (1 \pm i)$$

$$z: e^{\pm \pi/4} z^{1/2}$$

$$\overline{(z - e^{i\pi/4})(z - e^{-i\pi/4})} =$$

$$\frac{A}{z - e^{i\pi/4}} + \frac{B}{z - e^{-i\pi/4}} =$$

$$A(z - e^{-i\pi/4}) + B(z - e^{i\pi/4}) = 1$$

$$A + B = 0$$

$$-Ae^{i\pi/4} - Be^{-i\pi/4} = 1$$

$$Be^{i\pi/4} - Be^{-i\pi/4} = 1$$

$$B(\sin(\pi/4)) \frac{2i}{2} = 1$$

$$B = \frac{2i}{\sqrt{2}} = 1 \Rightarrow B = -\frac{\sqrt{2}}{2} i$$

$$A = \frac{\sqrt{2}}{2} i$$

$$\frac{\sqrt{2} i}{2} \int \left(\frac{z^{1/2}}{z - e^{i\pi/4}} - \frac{z^{1/2}}{z - e^{-i\pi/4}} \right) dz$$

$$= 2\pi i \left(\frac{\sqrt{2} i}{2} \right) \left(e^{i\pi/8} - e^{-i\pi/8} \right)$$

$$= \frac{1}{2} \sqrt{2} \sin \left(\frac{\pi}{4} \right)$$

$$(z - +)(z - -)$$

$$\frac{z^{1/2}}{(1)(1)} = \frac{A}{(1)} + \frac{B}{(1)}$$

$$Az = A e^{i\pi} + Bz$$

$$z^{1/2} \left(\frac{-e^{-i\pi} + e^{i\pi}}{(1)(1)} \right) \frac{\sqrt{2}}{2} i \frac{z_0}{z_0}$$

$$\sin(\pi/4) = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \left(\frac{\sqrt{2}(1-1)z'}{2} \right) = -1$$

$$z^2 + \sqrt{2}z + 1$$

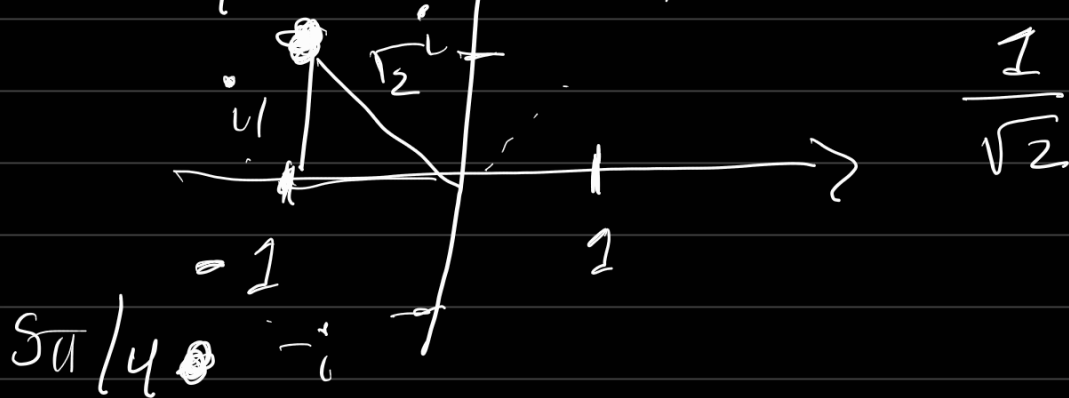
$$\frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

$$= \frac{1}{\sqrt{2}} (-1 \pm i)$$

$$\frac{1}{\sqrt{2}} (-1 + i) \quad ; \quad \frac{1}{\sqrt{2}} (-1 - i)$$

$$\frac{1}{\sqrt{2}} (-1 + i) \quad - \frac{1}{\sqrt{2}}$$

$$3\pi/4 \quad \uparrow$$



$$e^{i5\pi/4} = -\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$$e^{i3\pi/4} - e^{i5\pi/4}$$

$$-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

$$= \frac{i}{\sqrt{2}}$$

$$\begin{pmatrix} \sqrt{2} \\ 0 \\ i \end{pmatrix} \begin{pmatrix} e^{i3\pi/8} - e^{i5\pi/8} \end{pmatrix}$$

$$\cos(3\pi/8) \quad \cos(5\pi/8)$$

$$* \quad \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2} = \frac{\sqrt{2}}{2}(-1 \pm i)$$

$$\frac{1}{\sqrt{2}}(-1 \pm i) = e^{i3\pi/4} \quad \& \quad e^{i5\pi/4}$$

$$\frac{e^{i3\pi/8}}{e^{i5\pi/4} - e^{i\pi/4}} + \frac{e^{i5\pi/8}}{e^{i3\pi/4} - e^{i7\pi/4}}$$

$$= \frac{e^{i3\pi/8} - e^{i5\pi/8}}{e^{i3\pi/4} - e^{i\pi/8}}$$

$$-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} = \frac{2i}{\sqrt{2}} = \sqrt{2}i$$

$$2\pi i \left(\frac{e^{i3\pi/8} - e^{i5\pi/8}}{\sqrt{2}i} \right) = 2^{3/2} i \sqrt{2} \sin\left(\frac{3\pi}{8}\right) \checkmark$$

$$\int_C \frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz \quad C: \int_R + \int_G$$

$$\int_R \frac{z^{1/2}}{1+\sqrt{2}z+z^2} dz + \int_{-\pi}^{\pi} + \int_{\epsilon}^R + \int_{-\pi}^{\pi}$$

$\theta \rightarrow 0$

$$\theta = \pi$$

$$z = r e^{i\theta}$$

$$\int_R \frac{r^{1/2} e^{i\theta/2}}{1+\sqrt{2}r e^{i\theta} + r^2 e^{i2\theta}} dr$$

$$= \int_R \frac{r^{1/2}}{1+\sqrt{2}r+r^2} dr$$

$$\int_{\epsilon}^R \frac{r^{1/2} e^{-i\pi/2}}{1-\sqrt{2}r+r^2} dr + \int_R^{\epsilon} \frac{r^{1/2} e^{i\pi/2}}{1+\sqrt{2}r+r^2} dr$$

$$-i2 \int_{\epsilon}^R \frac{r^{1/2}}{1-\sqrt{2}r+r^2} dr = 2^{3/2} \pi i \sin\left(\frac{3\pi}{8}\right)$$

$$= -\pi \sqrt{2} \sin\left(\frac{3\pi}{8}\right)$$

$$-i \int_{\epsilon}^R \frac{r^{1/2}}{1-\sqrt{2}r+r^2} dr$$

$$\int_{\epsilon}^R \frac{(-r)^{1/2} (-i)}{1+\sqrt{2}r+r^2} dr$$

